ARJUNA

JEE 2026 BATCH

Chemistry

Chemical Equilibrium

Lecture No- 04

By – Rahul Dudi Sir

Topics to be covered

1. Expression of equilibrium constant
2. Degree of dissociation
3. Characteristics of equilibrium constant
4. Effect of temperature on equilibrium constant

Expression of Equilibrium Constant

Expression of Equilibrium Constant

Done

* Only reactants are present initially :→

Initial moles of Reactant → a
Vol. of Container → V
Total Pressure at eq. → P

* $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

	$PCl_5(g)$	$PCl_3(g)$	$Cl_2(g)$
$t=0$	a	0	0
$t=t_{sec}$	$a-x$	x	x

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{(x/V)(x/V)}{(a-x/V)}$$

$$K_c = \frac{x^2}{V(a-x)}$$

* Total moles at eq. :→ $a-x+x+x = a+x$

At eq. $x_{PCl_5} = \frac{a-x}{a+x}$ | $x_{PCl_3} = \frac{x}{a+x} = x_{Cl_2}$

$P_{PCl_5} = x_{PCl_5} \cdot P_T = \frac{a-x}{a+x} \cdot P$ | $P_{PCl_3} = \frac{x}{a+x} \cdot P = P_{Cl_2}$

$$K_p = \frac{P_{PCl_3} \cdot P_{Cl_2}}{P_{PCl_5}}$$

$$= \frac{\left[\frac{x}{a+x} \cdot P\right]\left[\frac{x}{a+x} \cdot P\right]}{\left[\frac{a-x}{a+x} \cdot P\right]}$$

Done

$$K_p = \frac{x^2 \cdot P}{a^2 - x^2}$$

Ans

Expression of Equilibrium Constant

* $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

$t=0$	a	0	0
$t=t_{eq}$	$a-x$	$\left(\frac{x}{2}\right)$	$\left(\frac{x}{2}\right)$

$$K_c = \frac{[H_2][I_2]}{[HI]^2} = \frac{\left(\frac{x}{2V}\right)\left(\frac{x}{2V}\right)}{\left(\frac{a-x}{V}\right)^2}$$

$$K_c = \frac{x^2}{4(a-x)^2}$$

Total moles at eq :-> $a-x+\frac{x}{2}+x/2$

$\Rightarrow a$

$$P_{HI} = \chi_{HI} \cdot P = \frac{a-x}{a} \cdot P$$

$$P_{H_2} = \chi_{H_2} \cdot P = \frac{x}{2a} \cdot P = P_{I_2}$$

$$K_p = \frac{P_{H_2} \cdot P_{I_2}}{(P_{HI})^2}$$

$$K_p = \frac{\left(\frac{x}{2a} \cdot P\right)\left(\frac{x}{2a} \cdot P\right)}{\left(\frac{a-x}{a} \cdot P\right)^2}$$

$$K_p = \frac{x^2}{4(a-x)^2}$$

$$K_p = K_c \ (\Delta n = 0)$$

Done

Expression of Equilibrium Constant

* $1N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$t=0$	a	b	0
$t=t_{eq}$	$(a-x)$	$(b-3x)$	$(2x)$

$K_c, K_p \rightarrow$ HW

Expression of Equilibrium Constant

Done

Total Pressure at eq. → P

* $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

t=0	a	0	0
t=teq.	a-x	x	x

$K_c = [CO_2] = \frac{x}{V}$

At eq. → $P_{CO_2} = X_{CO_2} \cdot P$

$= \frac{x}{(a-x+x+x)} \cdot P$

$P_{CO_2} = \frac{x}{x} \cdot P$

$P_{CO_2} = P$

$K_P = P_{CO_2} = P$ Ans.

Question

For a chemical reaction A + B $\rightleftharpoons$ 2C, 2 moles of A and 3 moles of B were taken initially. If $K_c = 4$ at 400 K, what will be the ~~value~~ moles of C at equilibrium?

Ans:→

$$A + B \rightleftharpoons 2C$$

	A	B	2C
t=0	2	3	0
t=teq.	2-x	3-x	2x

$$K_c = \frac{[C]^2}{[A][B]} = 4 = \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{2-x}{V}\right)\left(\frac{3-x}{V}\right)}$$

$$\cancel{4} = \frac{\cancel{4}x^2}{(2-x)(3-x)}$$

$$x = 6/5$$

moles of c at eq. :- 2x

$$= 2(6/5) = 12/5 \text{ Ans.}$$

Question

15 moles of H_2 and 5.2 moles of I_2 are mixed and then allowed to attain equilibrium at 500°C. At equilibrium, the concentration of HI is found to be 10 moles. The equilibrium constant for the formation of HI is

(A) 50

(B) 15

(C) 100

(D) 25

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$

	H_2	I_2	$2HI$
$t=0$	15	5.2	0
$t=t_{eq}$	$(15-x)$	$(5.2-x)$	$2x$
	$(15-5=10)$	$(5.2-5=0.2)$	

$2x = 10$

$x = 5$

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(10/V)^2}{\left(\frac{10}{V}\right)\left(\frac{0.2}{V}\right)}$$

$K_c = 50$ Ans

Question

HW

$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$,

For the reaction 1 mole each of acid and alcohol were taken initially at equilibrium. How many moles of ester will be formed ($K_c = 4$)

(A) 4/27

(B) 27/4

(C) 1/27

(D) 2/3

Question

HW

Equimolar conc. of H_2 and I_2 are heated to equilibrium in a $2l$ flask. At equilibrium the forward and backward rate constants are found to be equal. What percent of initial conc. of H_2 has reacted at equilibrium?

(A) 44%

(B) 66%

(C) 33%

(D) 20%

Question

1 mole of N_2 and 2 moles of H_2 are allowed to react in a 1 dm^3 vessel. At equilibrium 0.8 mole of NH_3 is formed. The concentration of H_2 in the vessel is

(A) 0.6 mol

(B) 0.8 mol

(C) 0.2 mol

(D) 0.4 mol

1L.

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

t=0	1	2	0
t=teq.	1-x	2-3x	2x.

$2x = 0.8$

$x = 0.4$

$2 - 3 \times (0.4)$

$\Rightarrow 2 - 1.2$

$= 0.8$

Question

HW

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$ (2022 Main, 27 June I)

In an experiment, 2.0 moles of NOCl was placed in a one-litre flask and the concentration of NO after equilibrium established, was found to be 0.4 mol/L. The equilibrium constant at 30°C is_____ $\times 10^{-4}$.

* Initially both reactants and Products are Present :->

Ex :-> $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$

$t = t_{sec}$	a	b	c	d
$t = t_{eq}$	$a-x$	$b-x$	$c+x$	$d+x$

when Rxn Proceed in forward direction.

Ex :-> $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$

when Rxn Proceeds in backward drn.

$t = t_{i}$	a	b	c	d
$t = t_{eq}$	$a+x$	$b+x$	$c-x$	$d-x$

Done

Degree of dissociation : (α)

$$A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$$

$t=0$	a	0	0
$t=t_{eq}$	$a-x$	x	x
	$(a-a\alpha)$	$a\alpha$	$a\alpha$

$$\alpha = \frac{\text{No. of moles Dissociated}}{\text{Total initial moles}}$$

$$\alpha = \frac{x}{a}$$

$$\% \alpha = \frac{x}{a} \times 100\%$$

$$x = a\alpha$$

Degree of dissociation :

α

* $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$A(g) \rightleftharpoons B(g) + C(g)$

$t=0$	a	0	0
$t=t_{eq}$	$a-x$	x	x
	$a-a\alpha$	$a\alpha$	$a\alpha$

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{a\alpha}{V}\right)\left(\frac{a\alpha}{V}\right)}{\left(\frac{a-a\alpha}{V}\right)}$$

* Eq. constant in terms of α :→

$$\boxed{K_c = \frac{a^2\alpha^2}{(a-a\alpha)V} = \frac{a\alpha^2}{(1-\alpha)\cdot V}}$$

→ $A(g) \rightleftharpoons B(g) + C(g)$

* Approximation :→

$\alpha <<< 1$, $1-\alpha \simeq 1$ (neglect)

$$K_c = \frac{a\alpha^2}{V}$$

Const. Const. → $\alpha^2 \propto V$

$$\alpha \propto \sqrt{V}$$

Degree of dissociation :

Total moles at eq :⇒ $a - a\alpha + a\alpha + a\alpha$

$\Rightarrow (a + a\alpha)$.

$$P_{PCl_3} = \chi_{PCl_3} \cdot P = \frac{a\alpha}{a+a\alpha} \cdot P = \left(\frac{\alpha}{1+\alpha}\right) \cdot P$$

$$P_{Cl_2} = \left(\frac{\alpha}{1+\alpha}\right) \cdot P$$

$$P_{PCl_5} = \chi_{PCl_3} \cdot P = \frac{a - a\alpha}{a + a\alpha} \cdot P$$

$$= \frac{1-\alpha}{1+\alpha} \cdot P.$$

Done

$$K_P = \frac{P_{PCl_3} \cdot P_{Cl_2}}{P_{PCl_5}}$$

$$= \frac{\left(\frac{\alpha}{\cancel{1+\alpha}}\right)^2 \cdot P^2}{\left(\frac{1-\alpha}{\cancel{1+\alpha}} \cdot P\right)}$$

$$K_P = \frac{\alpha^2 \cdot P}{1-\alpha^2}$$

→ To calculate the value of K_P, initial moles of reactant (a) are not required.

Approximation :⇒

$\alpha <<< 1$, $1 - \alpha^2 \simeq 1$.

$$\underset{\text{Const.}}{K_P} = \alpha^2 \cdot P$$

$$\alpha^2 \cdot P = \text{Const.}$$

$$\alpha^2 \propto \frac{1}{P}$$

$$\alpha \propto \frac{1}{\sqrt{P}}$$

* $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$] → $2AB(g) \rightleftharpoons A_2(g) + B_2(g)$.

→ $\Delta n = 0,\ K_c = K_p$

$t=0$	a	0	0
$t=t_{eq.}$	$a-x$	$x/2$	$x/2$
	$a-a\alpha$	$\frac{a\alpha}{2}$	$\frac{a\alpha}{2}$

$$K_c = \frac{[H_2][I_2]}{[HI]^2} = \frac{\left(\frac{a\alpha}{2V}\right)\left(\frac{a\alpha}{2V}\right)}{\left(\frac{a-a\alpha}{V}\right)^2}$$

$$K_c = \frac{a^2\alpha^2}{4(a-a\alpha)^2}$$

$$K_c = \frac{\alpha^2}{4(1-\alpha)^2}$$

Done

Approximation :→ $\alpha <<< 1$, $1-\alpha \approx 1$.

$$\underbrace{K_c}_{\text{Constant}} = \frac{\alpha^2}{\underbrace{4}_{\text{cons}}}$$

→ Const. at a Particular temp.

α = Constant

for Rx^n having ($\Delta n = 0$), α is constant at a Particular temperature. [α depends on temp. only]

* $A_{(g)} \rightleftharpoons 2B_{(g)}$.

$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

$K_c, K_p \rightarrow$ Home work.

In terms of α.

Question

$A(g) \rightleftharpoons B(g) + C(g)$

For the reaction,
$N_2O_3(g) \rightleftharpoons NO(g) + NO_2(g)$: total pressure = P, degreee of dissociation = 50%. Then K_p would be:

(A) 3 P

(B) 2P

(C) P/3

(D) P/2

M#1

$\% \alpha = 50\%$

$\alpha = 0.5$

$K_p = ?$

$K_p = \frac{\alpha^2}{1-\alpha^2} \cdot P$

$K_p = P/3$

M#2

$N_2O_3(g) \rightleftharpoons NO(g) + NO_2(g)$

t=0	1 (a)	0	0
t=teq.	$1-\alpha$ (a−aα) (0.5)	α (0.5)	α (0.5)

$\chi_{N_2O_3} = \frac{0.5}{1.5}$

$P_{N_2O_3} = \frac{0.5}{1.5} \cdot P = P/3$

$P_{NO} = P/3$

$P_{NO_2} = P/3$

$K_p = \frac{P_{NO} \cdot P_{NO_2}}{P_{N_2O_3}}$

$K_p = P/3$

Question

HW

For the reaction,

$2AB(g) \rightleftharpoons A_2(g) + B_2(g)$

Equilibrium constant is 1/64 at 444 K. the degree of dissociation of AB will be:

$\alpha = ?$

(A) 10 %

(B) 20 %

(C) 30 %

(D) 40 %

Ans: [B]

Characteristics of Equilibrium Constant

Characteristics of Equilibrium Constant

Done

- Expression for equilibrium constant is applicable only when concentrations of the reactants and products have attained constant value at equilibrium state.
- The value of equilibrium constant is independent of initial concentrations of the reactants and products.
- Equilibrium constant is temperature dependent having one unique value for a particular reaction represented by a balanced equation at a given temperature.

$K_{eq} = \frac{K_f}{K_b}$

Only temp. dependent.

$K_f, K_b \rightarrow$ Temperature dependent only.

Characteristics of Equilibrium Constant

- **The equilibrium constant for the reverse reaction is equal to the inverse of the equilibrium constant for the forward reaction.**

Done

$$A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}, \quad K_c = \frac{[B][C]}{[A]}$$

$$B_{(g)} + C_{(g)} \rightleftharpoons A_{(g)}, \quad K_c' = \frac{[A]}{[B][C]}$$

$$K_c' = \frac{1}{K_c}$$

Characteristics of Equilibrium Constant

Done

- **When a chemical reaction is multiplied by a factor n**

$A(g) \rightleftharpoons B(g) + C(g)$, $K_c = \frac{[B][C]}{[A]}$

$2A(g) \rightleftharpoons 2B(g) + 2C(g)$. $K_c' = \frac{[B]^2[C]^2}{[A]^2}$

$K_c' = (K_c)^2$

$\frac{1}{2}A(g) \rightleftharpoons \frac{1}{2}B(g) + C(g)$

$K_c'' = (K_c)^{1/2}$

Characteristics of Equilibrium Constant

- **When a chemical reaction is written in more than one steps-**

$A_{(g)} \rightleftharpoons B_{(g)}$, $K_1 = [B]/[A]$

$+$

$B_{(g)} \rightleftharpoons C_{(g)}$, $K_2 = [C]/[B]$

$+$

$C_{(g)} \rightleftharpoons D_{(g)}$, $K_3 = [D]/[C]$

$A_{(g)} \rightleftharpoons D_{(g)}$, $K = \frac{[D]}{[A]}$

$K = K_1 \times K_2 \times K_3$

$$* \; H_3PO_4 \rightleftharpoons 3H^+ + PO_4^{3-}, \; K$$

$$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-, \; K_1$$

$$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}, \; K_2$$

$$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}, \; K_3$$

$$K = K_1 \times K_2 \times K_3$$

Question

If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the Equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be,

(A) $K^{\frac{1}{2}}$

(B) $\frac{1}{2}K$

(C) K

(D) K^2

Question

The equilibrium constant of the following are:

$N_2 + 3H_2 \rightleftharpoons 2NH_3 \longrightarrow K_1$

$N_2 + O_2 \rightleftharpoons 2NO \longrightarrow K_2$

$H_2 + \frac{1}{2}O_2 \rightleftharpoons H_2O \longrightarrow K_3$

The equilibrium constant (K) of the reaction:

$2NH_3 + \frac{5}{2}O_2 \rightleftharpoons 2NO + 3H_2O$

(A) $K_2K_3^3/K_1$

(B) K_2K_3/K_1

(C) $K_2^3K_3/K_1$

(D) $K_1K_3^3/K_2$

$2NH_3 \rightleftharpoons N_2 + 3H_2(g)$, $1/K_1$

$N_2 + O_2 \rightleftharpoons 2NO$, K_2

$3H_2 + \frac{3}{2}O_2 \rightleftharpoons 3H_2O$, $(K_3)^3$

$\rightarrow \frac{K_2 \cdot K_3^3}{K_1}$

Question

HW

At 298 K (2023 Main, 29 January II)

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $K_1 = 4 \times 10^5$

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $K_2 = 1.6 \times 10^{12}$

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$, $K_3 = 1.0 \times 10^{-13}$

Based on above equilibria, the equilibrium constant of the reaction,

$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$

is ____ $\times 10^{-23}$ (Nearest Integer)

Ans: [4]

Question

HW

For the following reactions, equilibrium constants are given:

$S(s) + O_2(g) \rightleftharpoons SO_2(g)$; $K_1 = 10^{52}$

$2S(s) + 3O_2(g) \rightleftharpoons 2SO_3(g)$; $K_2 = 10^{129}$.

The equilibrium constant for the reaction,

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is (2019 Main, 8 April II)

(A) 10^{25}

(B) 10^{77}

(C) 10^{154}

(D) 10^{181}

Question

HW

Consider the following reversible chemical reactions,

$$A_2(g) + B_2(g) \overset{K_1}{\rightleftharpoons} 2AB(g)$$

$$6AB(g) \overset{K_2}{\rightleftharpoons} 3\,A_2(g) + 3\,B_2(g)$$

The relation between K and K, is (2019 Main, 9 Jan II)

(A) $K_2 = K$

(B) $K_1K_2 = 3$

(C) $K_2 = K_1^{-3}$

(D) $K_1K_2 = \frac{1}{3}$

Characteristics of Equilibrium Constant

- **Factors which do not influence equilibrium constant:**

1. **Concentration of reactants and products**
2. **Pressure and volume**
3. **Presence of catalyst**
4. **Addition of inert gas at constant pressure and volume**

Characteristics of Equilibrium Constant

- **Factors which influence equilibrium constant:**

1. **Mode of representation of chemical reaction**
2. **Stoichiometry of reaction**
3. **Temperature**

Aaj ke liye
itna hi
Physics Wallah

Homework from Module

- DPP.
- Revision.
- जो class में दिया है.

THANK *You*